

Connectivity Hitting in the Random Graph Process: Exact Finite Laws and the Gumbel Window

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Abstract

Reveal the $K = \binom{n}{2}$ edges of the complete labeled graph in uniform random order, and let τ_{conn} be the first connected prefix. Decomposition by the component containing vertex 1 gives an exact recurrence for every connected-graph count, hence the complete finite CDF, PMF, tail, support, and all moments of τ_{conn} .

1 The monotone process and connected counts

Let $e_1, \dots, e_K$ be a uniformly random permutation of the edges of K_n , and put $G_m = (\{1, \dots, n\}, \{e_1, \dots, e_m\})$. Thus G_m is uniform over the $\binom{K}{m}$ labeled graphs with m edges. Edge addition is monotone, and the connected graphs form an absorbing upper set. We take $\tau_{\text{conn}} = \min\{m : G_m \text{ is connected}\}$, with $\tau_{\text{conn}} = 0$ when $n = 1$.

Write $C(n, m)$ for the number of connected labeled graphs on n vertices with m edges. Set $C(1, 0) = 1$, and interpret both $C(s, j)$ and binomial coefficients as zero outside their natural ranges.

Theorem 1 (All finite connectivity-hitting laws). *For $n \geq 2$, $K = \binom{n}{2}$, and every integer m ,*

$$C(n, m) = \binom{K}{m} - \sum_{s=1}^{n-1} \binom{n-1}{s-1} \sum_j C(s, j) \binom{\binom{n-s}{2}}{m-j}. \quad (1)$$

Moreover $C(n, m) = 0$ for $m < n - 1$,

$$C(n, n-1) = n^{n-2}, \quad C(n, K) = 1. \quad (2)$$

For $0 \leq m \leq K$, with $F_n(-1) = 0$,

$$F_n(m) := \mathbb{P}(\tau_{\text{conn}} \leq m) = \frac{C(n, m)}{\binom{K}{m}}, \quad (3)$$

$$\mathbb{P}(\tau_{\text{conn}} = m) = F_n(m) - F_n(m-1), \quad (4)$$

$$\mathbb{P}(\tau_{\text{conn}} > m) = 1 - F_n(m). \quad (5)$$

For every integer $r \geq 1$,

$$\mathbb{E}[\tau_{\text{conn}}^r] = \sum_{m=0}^{K-1} ((m+1)^r - m^r) \{1 - F_n(m)\}. \quad (6)$$

Finally, for $n \geq 2$,

$$n-1 \leq \tau_{\text{conn}} \leq \binom{n-1}{2} + 1. \quad (7)$$

Proof. If a graph is disconnected, let s be the size of the component containing vertex 1. Choose its other $s-1$ labels, choose its connected j -edge subgraph, and place every remaining edge among the other $n-s$ vertices. This decomposition is unique and subtracting it from all m -edge graphs gives (1). A connected graph needs at least $n-1$ edges; at equality it is a tree, counted by the Prüfer-code value n^{n-2} . The complete graph is unique, proving (2).

The first m positions of a uniform edge permutation form a uniform m -subset, which proves (3). Monotonicity gives (4)–(5). For every nonnegative integer-valued T , $T^r = \sum_{m=0}^{T-1} ((m+1)^r - m^r)$; expectation proves (6). The smallest connected graph is a tree. A disconnected graph has at most $\binom{n-1}{2}$ edges, since convexity maximizes the sum of within-component edge capacities at component sizes $n-1$ and 1. This proves (7). $\square$

Source ownership and AI-use statement

Erdős and Rényi [1] established the random-graph evolution and connectivity threshold. We claim no priority for that result. The present package supplies a self-contained frozen-model derivation, exact finite hitting atlas, and auditable boundary ledger. AI tools assisted symbolic checks, hostile mutation design, and manuscript preparation; all proof obligations, cutoffs, and nonclaims are explicit.

References

- [1] P. Erdős and A. Rényi, “On the evolution of random graphs,” *Publications of the Mathematical Institute of the Hungarian Academy of Sciences* 5 (1960), 17–61.